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Influence of trace tungsten contents on thin-film properties and electromigration behaviors of electroless-deposited cobalt interconnect lines

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ABSTRACT

Electroless-plated cobalt alloys with tungsten typically exceeding 5 at.% are the common barrier materials for copper metallization. As cobalt is a new interconnect material for advanced microchips, it is worth studying the advanced cobalt-interconnect metallization process that involves extremely diluted cobalt-tungsten (CoW) alloys with an eliminated impurity resistivity and enhanced electromigration reliability. To this aim, we fabricated three different CoW nanolines with 0.03, 0.06 and 0.11 at.% of W using an electroless process, and examined their electromigration behaviors under accelerated stressing conditions, using Co lines as a control. All three W contents, especially 0.06 at.%, mitigate Co electromigration. The mechanism for the electromigration mitigation is due to an improvement of the film's nanomechanical and surface wetting properties, as well as grain-structure refinements. A full account of the relationship between the film properties, grain structures and electromigration testing results is given, reasonably explaining the cause of the reliability optimization.

1. Introduction

The contact critical dimension of the CMOS transistors currently using copper (Cu) as a core interconnect material is expected to reach 18 nm [1], associated with a diminishment of interconnect width towards less than a few tens of nm. The rather long electron mean free path of 40 nm for Cu would induce multiple electron scatterings in such narrow lines, giving rise to a high residual electrical resistivity [2]. Therefore, Cu is not a good interconnect conductor for future technology nodes. Compared to Cu, cobalt (Co) has a much shorter electron mean free path of about 9 nm, leading to the product of the resistivity and mean free path of Co being less than that of Cu: 6.05×10^{-16} vs. $6.70 \times 10^{-16}\ \Omega\ m^2$ [2]. Considering the resistivity-size effect, Co is considered to be a more appropriate interconnect material than Cu, especially as the line width (or cross section) is reduced down to below a crossover value of about 15 nm [3–5] (or 450 nm^2) [6,7]. Moreover, electromigration is essentially a forced metal diffusion behavior and the key factor that influences the electromigration reliability of a conductive line is the magnitude of its effective valence (Z*) and diffusivity (D): the lower the product of the Z* and D, the higher the electromigration reliability. The absolute value of the Z* for Co is about 5 to 10 times lower than that of Cu [8,9]. With a higher melting point of 1495 °C than 1085 °C of Cu, Co has a much lower (about one order of magnitude) self-diffusivity than Cu [10]. Indeed, the product of the Z* and D for Co is two orders of magnitude smaller than that of Cu [11,12]. Therefore, Co has become the new, alternative interconnection material for advanced microchips [1].

Electroless process has long been used to prepare thin barrier/cap layers of Co alloys for Cu-interconnect metallization [13]. The basic components in an electroless cobalt-alloy plating solution are typically cobalt sulfate ($CoSO_4$) as a source of Co ions and dimethylamine borane (C_2H_9BN), sodium hydroborate ($NaBH_4$) or sodium hypophosphite (NaH_2PO_2) as a chemical reductant. The oxidation of the chemical reductant results in the incorporation of boron or phosphors into the deposited Co, creating a solute-strengthening effect that stabilizes the phase structure, stuffs grain boundaries and/or generates an amorphous matrix, which are all beneficial to barrier/cap properties. Moreover, as a refractory metal, tungsten (W) is highly resistant to heat and abrasion. Its addition to the binary Co alloys thus further elevates

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thermal-mechanical and barrier properties. Therefore, innovate formulations based on sulfate baths with an added W-containing salt, such as tungstate [14,15] or phosphotungstic acid [16], have been developed to fabricate barrier/cap layers of ternary Co alloys with high W contents usually exceeding 5 at.%.

To serve as a core-conducting interconnect material, thin metal films need to have an electrical resistivity close to their bulk values. However, high concentrations of alloy components, particularly heavy elements such as W, in thin films of binary or ternary cobalt-based alloys would induce a high impurity electrical resistivity. After thermal annealing, the tungsten-added cobalt binary/ternary films still give a high resistivity value exceeding several tens or even hundreds of μΩ-cm [17,18]. We have recently developed an all-wet self-assembled-monolayer (SAM) seeding and electroless process for cobalt metallization. By using a $CoSO_4$–N_2H_4 (hydrazine; a metal-free reductant) bath, thin films and nanolines of alloy-free Co with (after annealing) a low resistivity of ≤ 15 μΩ-cm are fabricated [19]. In this study, we report an extension of our previous work by adding tungstate to the $CoSO_4$–N_2H_4 bath for the interconnect metallization of various Co alloys with trace W contents (termed CoW) of 0.03, 0.06 and 0.11 at.%, giving a low impurity-component resistivity potentially suitable for future interconnect-material applications. By using unalloyed Co as a reference, the enhancement of the electromigration reliability of interconnect lines of the various CoW alloys, in particular with 0.06 at.% added W, will be confirmed. The correlation between nanomechanical/wetting properties of the surface films and the difference in electromigration failure profiles and grain structures of the interconnect lines will be analyzed, elucidating why the optimal W content (0.06 at.%) gives the greatest effectiveness of mitigating electromigration failure.

2. Experimental procedures

The experiments were designed to develop a new electroless process for the plating and thin-film properties characterization of extremely diluted Co alloys with various W contents of 0.03, 0.06 and 0.11 at.%. Then, interconnect lines of the three tungsten-added Co alloys embedded in patterned SiO_2 dielectric layers were fabricated for the evaluation of their electromigration behaviors, using interconnect lines of unalloyed Co as a control.

2.1. Substrate fabrication

Electron beam lithography in conjunction with plasma etching was used to pattern a 130 nm-thick SiO_2 layer on an 8-inch (100) Si wafer with groups of straight trenches, terminated by two squared recessions. The patterned SiO_2/Si wafer was cleaved into 2×2 cm^2 pieces for the fabrication of interconnect lines of the three different Co alloys and Co (the control sample for electromigration testing). In addition, a 30 nm-thick SiO_2 layer was thermally grown on a planar Si wafer. The planar SiO_2/Si wafer was also cleaved into 2×2 cm^2 pieces for the fabrication of thin Co and W-added Co films.

2.2. Surface-modification process and electroless metallization

Prior to the fabrication of electromigration testing structures and thin planar films of Co alloys and Co, the patterned and planar SiO_2/Si substrates were both processed by the same procedure: surface hydroxylation and the growth of a (3-aminopropyl)triethoxysilane SAM (APTMS-SAM), followed by SAM functionalization and seeding (catalytic particle growth). Silanization of the dielectric substrates with an orderly APTMS-SAM for the seeding was performed under a tightly controlled processing environment and in a suitable toluene-APTMS incubation solution, as detailed in our previous publications [20,21]. The readers may also refer to our previous publication for surface characterizations of the SAM-coated and seeded samples [22]. Analytical reagent grade chemicals and 18 mega-ohm deionized water were used for the preparation of the chemical solutions.

The planar SiO_2/Si substrates after seeding were deposited with a thin Co film as a control sample, using a recently developed $CoSO_4$ (0.2 M)–N_2H_4 bath formulation [19]. Na_2WO_4 with a low molar concentration of 2.50×10^{-4} M, 1.06×10^{-3} M or 5.30×10^{-3} M was added to the sulfate solution for the plating of W-containing Co films. Chemical composition analyses using Thermo Scientific inductively coupled plasma mass spectrometry (iCAP TQ) revealed that the cobalt-based films deposited by the above three baths contained 0.03, 0.06 and 0.11 at.% of W, denoted hereafter as CoW_l, CoW_m and CoW_h, respectively.

The patterned SiO_2/Si substrates after seeding were divided into four sets for the electroless filling of Co, CoW_l, CoW_m and CoW_h to a thickness of about 150 nm. After removing the planar-field deposits using mechanical polishing, numerous interconnect lines of Co, CoW_l, CoW_m and CoW_h were formed. The top-view scanning electron image (Fig. 1), together with the inserted cross-sectional transmission electron image, indicates that the electromigration test structure consists of a straight-line segment of 100 nm deep, 60 nm wide (the bottom side) and 25 μm long, which is terminated by two 30×30 μm^2 electrode pads.

2.3. Electromigration testing and failure analyses

Prior to electromigration testing, the samples of Co, CoW_l, CoW_m and CoW_h lines were annealed (400 °C/30 min) using an Ulvac's Mila-5000 rapid thermal vacuum annealer. All annealing treatments were performed in flowing Ar–H_2 (5 %) that prevented the samples from oxidation. A heating rate of 15 °C s^{-1} was used and fast cooling was performed by simply switching off the heating. They were then placed on a thermal chuck situated in a probe-station-enclosed dark box, immediately under constant current stressing by using a Keithley 4200-SCS semiconductor parameter analyzer. The onset of electromigration for metallic interconnects under regular operation conditions occurs on a time scale of several years. Here, the interconnect lines were tested under accelerated electromigration conditions using very high current densities of the order of 10^7 A cm^{-2}. Such current densities were about 2–3 orders of magnitude greater than the practical ones. Thus, the electromigration failure of the tested interconnect lines was speeded up. Specifically, many Co, CoW_l, CoW_m and CoW_h lines were stressed at room temperature under a preset current density of 30, 50 or 70 MA cm^{-2}, simultaneously recording the resistance versus stressing time plots until line reupture. Then, the mean times to failure (MTFs) of these lines were derived, used to calculate the current density scaling factor (or termed n factor) for each line type, based on modified Black's equation [23]:

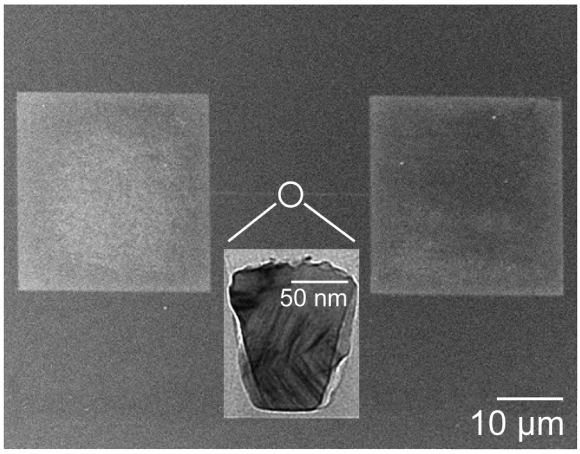


Fig. 1. Top-view SEM micrograph and cross-sectional TEM micrograph (inset) showing the electromigration test structure of Co and W-added Co interconnect lines.

$$\mathrm{MTF} = A\, J^{-n}\exp(E_a / kT), \quad (1)$$

where A is the cross-sectional dependent constant, J is the current density, E_a is the activation energy, k is the Boltzmann constant, and T is the absolute temperature.

Physical failure profiles and grain structures of the Co and CoW lines after electromigration stressing were analyzed by Thermo Scientific Talos™ F200X. This instrument was a (scanning) transmission electron microscope, (S)TEM, having multiple operating modes including conventional TEM imaging, scanning TEM (STEM) imaging, diffraction, spectroscopy and combinations of these. Particularly, it had an extremely high spatial resolution of 0.13 nm for high-resolution imaging and energy-dispersive X-ray (EDX) mapping. Thin-foil specimens for the STEM/TEM analysis were prepared by a Thermo Scientific Helios G5 DualBeam, equipped with a scanning electron microscope (SEM) for the *in-situ* cutting of the lines along their longitudinal directions using a focused ion beam.

2.4. Thin-film property characterization

Film thicknesses of Co, CoW_l, CoW_m and CoW_h were measured by a high-precision surface-step profilometer (ET 3000, Kosaka Lab. Ltd.), and their deposition rates were procured from the plots of film thickness versus plating time. The planar films for property examinations had the same thickness of about 70 nm. SEM imaging was used to observe the thermal stability of these films after annealing.

Nanomechanical scratch testing of thin films of Co, CoW_l, CoW_m and CoW_h was conducted in a Bruker's Hysitron TI 980 Tribolndenter. The Tribolndenter was equipped with two sensors for precise force and depth measurements using a state-of-the-art piezoelectric actuator. This advanced design allowed us to control each and every measurement by progressively increasing the normal load of a diamond stylus in a controlled fashion, while synchronously collecting the feedback data of normal displacement, lateral force and lateral displacement. Subsequently, the related plots were automatically generated by using a Hysitron TriboScan software. On-site scanning probe microscopical imaging of the whole scratch profile for the feedback of the scratch data was used to derive the nanomechanical properties of the tested films, such as critical load, delamination force, adhesion strength and friction coefficient.

Surface energies of SAM-coated SiO_2 (SAM/SiO_2) substrates and thin layers of Co, CoW_l, CoW_m and CoW_h on the SAM/SiO_2 substrates were measured by the Owens-Wendt method [24]. The measured surface energies were used to estimate the effect of the added-W content on the wettability and adhesion strength of the coated films.

3. Results and discussion

In this section we first describe the basic properties of the thin deposited films of Co, CoW_l, CoW_m and CoW_h, including their deposition rates, chemical compositions, surface-deposit evolutions and thermal stability, and then evaluate interconnect's electromigration behaviors and electromigration-mitigation mechanisms. Finally, a summary and conclusions are presented.

3.1. Basic film properties

Thin films of Co, CoW_l, CoW_m and CoW_h were deposited on the same seeded SiO_2/Si substrates for various times. Then, their thicknesses were measured by high-resolution surface profilometry (some also reconfirmed by top-view SEM imaging). It was found that the deposition of these films with thicknesses in the range of 30 nm–100 nm was reproducible and the deposited layers were bright colored and uniform under visual inspection. Least-squares fitting the thickness versus deposition time plots in Fig. 2 results in four straight-line segments, giving a constant deposition rate for each kind of the four deposits. It is clear from

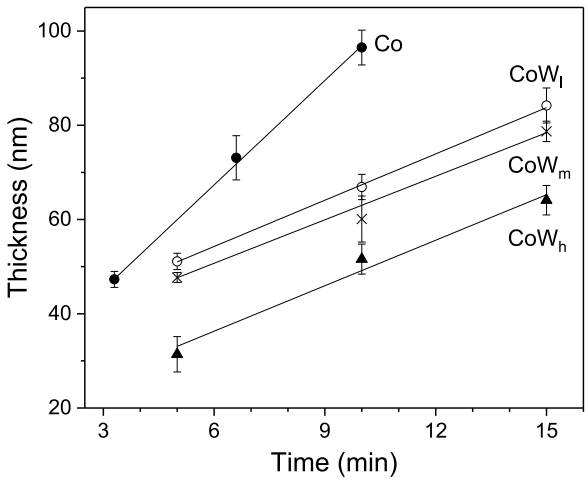


Fig. 2. Plots showing the linear variations of thickness with plating time for thin films of Co, CoW_l, CoW_m and CoW_h, from which their deposition rates were obtained by least squares fitting to be 7.80 5.58, 5.22 and 4.20 nm min^{-1}, respectively.

Fig. 2 that the Co films, deposited by the cobalt sulfate-citrate-hydrazine bath, have the highest deposition rate of 7.80 nm min^{-1}. Here, the Co films were deposited by a hydrazine-based bath with 0.2 M cobalt sulfate. When tungstate with increasing concentrations from 0.25, 1.06 to 5.30 mM is respectively added to the above electrolyte as a W source, the deposition rates of the thin CoW_l, CoW_m and CoW_h films are monotonically decreased to be 5.58, 5.22 and 4.20 nm min^{-1}, respectively.

When adsorbed onto the deposit surface, W (VI) ions tend to become an inhibiting component that blocks the self-catalytic sites of Co [25]. Thus, the W (VI) ions depress Co deposition and their reduction reaction into metallic W being incorporated to the main Co deposit is interdependent upon the main reduction reaction of the cobaltous ions. Specifically, previous studies have shown that adding high contents of tungstate (WO_3^{4+}) to the cobalt sulfate (Co^{2+}) aqueous solutions with WO_3^{4+}–to–Co^{2+} molarity ratios exceeding 0.3 or even greater than 2, using a given chemical reductant such as hypophosphite or dimethylamine borane, results in a decrease in the deposition rates of Co [25–28]. Here, the pure Co film was deposited by a sulfate-citrate-hydrazine bath. It should be noted that the concentration of the sulfate in this bath was 0.2 M. The CoW_l, CoW_m and CoW_h films were deposited by otherwise this same bath except including quite low tungstate concentrations of about 2.50×10^{-4}, 1.06×10^{-3} and 5.30×10^{-3} M, respectively. The WO_3^{4+}–to–Co^{2+} molarity ratios of the three CoW electroless-plating baths were very low, only 1.20×10^{-3}, 5.30×10^{-3} and 2.65×10^{-2}. However, they do cause the Co deposits to yield gradually decreased deposition rates. These results suggest that the tungstate in the tungstate-diluted Co-plating baths also depresses Co deposition as it does in the tungstate-concentrated baths.

Thin films of binary cobalt alloys, such as Co(P) or Co(B), are typically deposited by aqueous electrolytes that contain cobalt sulfate, hypophosphite (or borazane) and a citrate chelating agent. Tungstate, as used in this study, is usually added to the above electroless baths for the incorporation of metallic W into the binary alloys for barrier/cap applications. Traditionally, highly concentrated tungstate, in terms of tungstate-to-cobalt sulfate (or WO_3^{4+}–to–Co^{2+}) molarity ratios of 0.3 or greater, is used, which results in the incorporation of substantial amounts of metallic W in the range of 2–10 at.% to the Co matrix. In spite of the fact that pure W cannot be deposited by either chemical or electrochemical methods, electrochemical studies suggest that Co serves as a catalyst for the electroless deposition of W; namely, after the deposition of Co on catalytic particles or layers, the deposited Co subsequently serves as another catalyst to activate the codeposition of Co and W. The W can be reduced from tungstile ions (WO_2^{2+}), and is then

simultaneously incorporated into the Co matrix during the reduction of the cobaltous ions via a so-called induced codeposition mechanism [16,28,29]. Here, the baths with extremely low tungstate–to–cobalt sulfate molarity ratios of 1.20×10^{-3}, 5.30×10^{-3} and 2.65×10^{-2} were used to deposit CoW alloys, and the deposited films indeed contained only 0.03, 0.06 and 0.11 atomic percentage of W, respectively. This result suggests that, for the cobalt-tungsten alloy films deposited by the tungstate-diluted hydrazine baths, the trace W contents are incorporated to the Co film matrixes also via the induced codeposition mechanism, as for the conventional tungstate-concentrated baths. Indeed, the study of the deposition and film properties of tungsten-containing binary/ternary alloys of Co using $CoSO_4$–N_2H_4 bath formulations with diluted tungsten-ion sources is rather limited. The mechanism for plating diluted cobalt-tungsten alloys using the tungstate-diluted baths is worthy of investigation via electrochemical analysis. This topic is behind the scope of this current study, and a detailed electrochemical investigation will be made in a separate project which will be presented elsewhere.

The seeding process used in this study is capable of loading 2-nm-sized catalytic particles onto the substrates of SAM-coated SiO_2/Si [22], which are the catalytic sites for the electroless growth of W-alloyed Co or unalloyed Co. During the course of electroless deposition, the W-added Co or Co deposits surrounding the 2 nm-sized catalytic particles are expanded with plating time to become island-like deposits, eventually coalescing themselves into a thin continuous film. As demonstrated by the top-view and cross-sectional (inset) SEM micrographs in Fig. 3a and 3b, respectively, the thin Co and CoW_m films yield similar uniform surface morphologies and microstructures. The two Co and CoW_m films were further annealed to a certain extent (500 °C/30 min in flowing Ar/H_2 gas), and the associated top-view SEM micrographs are shown in Fig. 3c and 3d, respectively. It can be seen from Fig. 3c that the annealing causes a serious disruption and agglomeration of the Co film, leading to the exposure of the underlying dielectric substrate. However, an identical annealing induces densification and substantial grain growth of the CoW_m film, which still retains an intact film structure (Fig. 3d). These findings suggest that the Co film is strengthened by the extremely diluted alloying of 0.06 at.% W.

3.2. Electromigration performance

A set of pristine Co, CoW_l, CoW_m and CoW_h interconnect lines was stressed at ambient temperature for various durations under 30 MA cm^{-2}, simultaneously their resistance versus time plots were recorded, as shown in Fig. 4a. The related plots for two other sets of pristine lines under two other different current densities (50 and 70 MA cm^{-2}) were recorded, displayed in Fig. 4b and c, respectively. We can point out the existence of two regions for all resistance-changing curves: first, the resistances remain very close to the initial resistance values and then leap at a given stressing time, corresponding to the onset of line failures. The time when a rupture of the lines occurs corresponds to a sharp drop in the electrical resistance, defined as the electromigration failure time. Five meaningful resistance versus stressing-time plots for each line type under the same stressing condition were measured. The obtained electromigration failure times were averaged to obtain the MTF. Fig. 5 shows that the logarithmic plots of the MTFs versus current densities for the four different types of metallic interconnects constitute four straight-line segments, corresponding to the current density scaling factors (n factors) of 0.50, 0.43, 0.33, 0.42 for the Co, CoW_l, CoW_m and CoW_h lines, respectively.

While not applicable to the entire interconnect routes with changes in current flow directions or layered structures, Black's equation, Eq. (1), fits quite well for linear interconnects, and is thus an accurate way of estimating the service life duration and failure mode (through the magnitude of current-density scaling factor) of simple interconnect lines under accelerated testing conditions [30,31]. Previous studies on electromigration behaviors of the conductive lines of Al-Cu and Cu by Lloyd among others [23,32] came to a conclusion that the magnitude of the scaling factor provides an insight into the electromigration failure mode of metallic lines. When sealed by an adhesive/barrier, the conductive line normally exhibits a void-growth failure mode with a scaling factor of 1 or less, depending on the robust of the sealant layer and the mechanical strength of the core metallic material itself. Here, the interconnect lines of Co, CoW_l, CoW_m and CoW_h are sealed by an SAM barrier and all reasonably give a scaling factor of ≤ 0.5 over the accelerated electromigration stressing currents from 30 to 70 MA cm^{-2}. These results suggest the failure of all these lines by a void-growth mode.

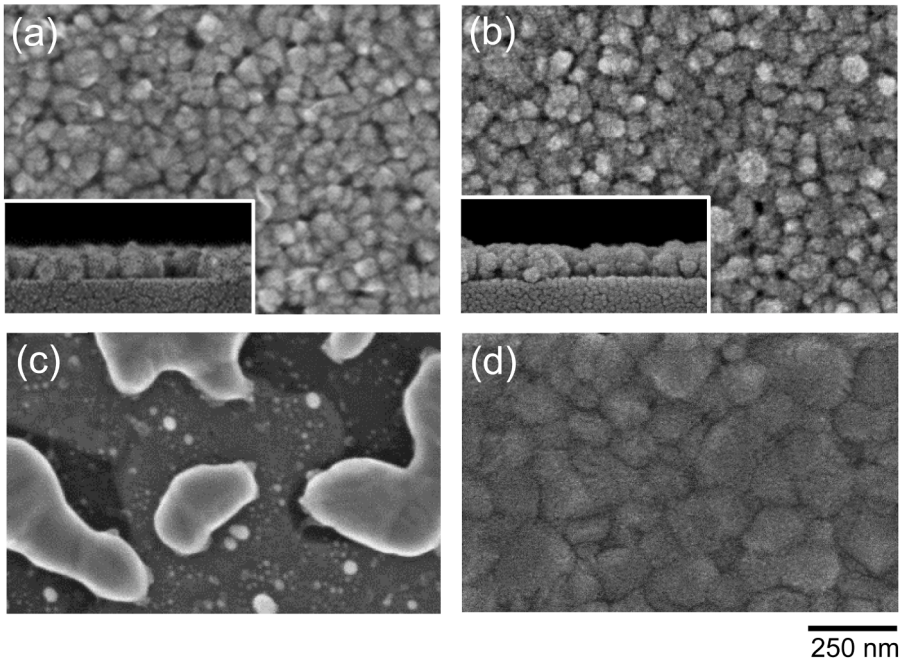


Fig. 3. Top-view and cross-sectional (inset) SEM micrographs typical of as-grown (a) Co and (b) CoW_m films, and the associated SEM micrographs for the (c) Co and (d) CoW_m films after 500 °C/30 min inert-gas annealing.

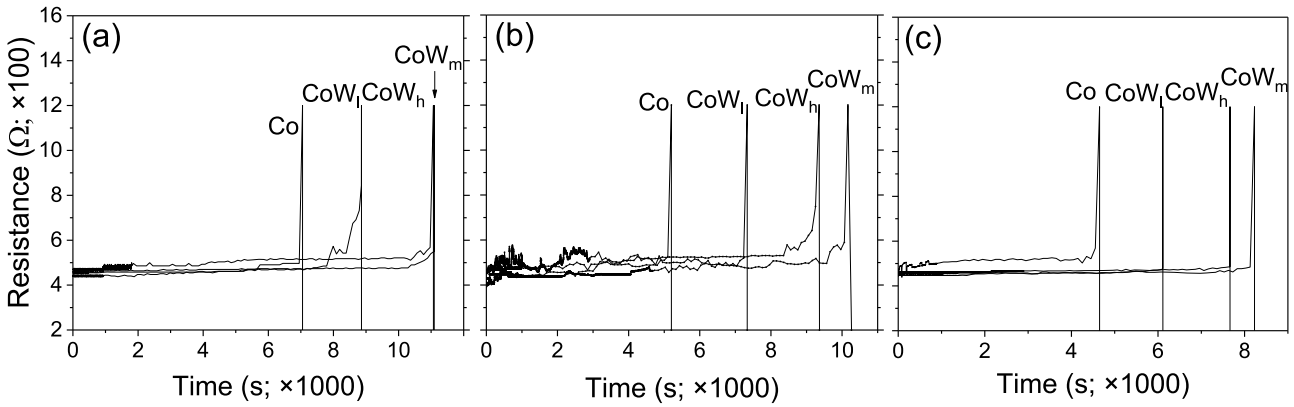


Fig. 4. Resistance against stressing time curves for the pristine lines of Co, CoW$_l$, CoW$_m$ and CoW$_h$ bias-stressed at 25 °C under (a) 30, (b) 50 and (c) 70 MA cm^{-2}.

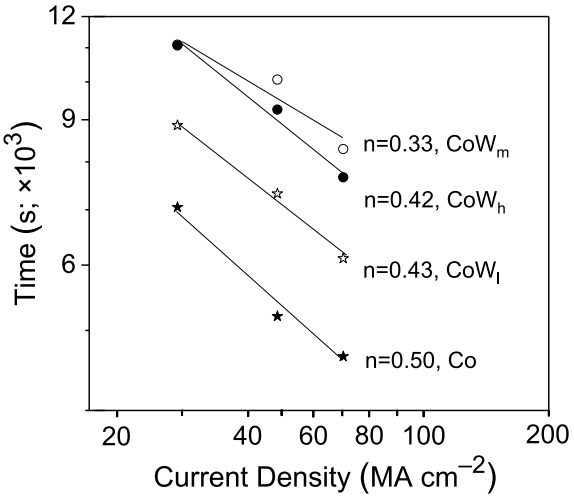


Fig. 5. Logarithmic plots of electromigration failure time against current density for Co and various W-added Co interconnect lines. The straight-line segments were obtained by least squares fitting, from which the scaling factors were extracted and shown in the figure.

The Co line has the shortest lifetime and highest scaling factor of 0.50, indicating that it has the fastest void growth rate and is most liable to electromigration failure. All other three W-alloyed Co lines have a lower scaling factor and longer lifetime than the Co line, suggesting the retarding of void growth by the added W. Especially, the CoW$_m$ (W = 0.06 at.%) line has the lowest scaling factor of 0.33. This result suggests that the medium W content of 0.06 at.% gives the best alloying effect for the CoW$_m$ line to yield the lowest void growth rate upon electromigration stressing.

3.3. Thin-film properties

The samples for nanoscratch testing were thin films of Co, CoW$_l$, CoW$_m$ and CoW$_h$ on the same SAM-tethered SiO_2 on Si. These films had the same thickness of 70 nm. Their nanomechanical properties were measured by nanoscratch testing using a diamond stylus to draw across the film surfaces at a constant speed (0.6 μm s^{-1}) with progressively increased normal forces towards a maximum predetermined force. Considering the difference in film hardness and adhesion strength, the normal force was increased at a constant rate of 10 μN s^{-1} and 15 μN s^{-1} for the Co/CoW$_l$ films and the CoW$_m$/CoW$_h$ films, respectively. The typical sets of the plots of lateral force versus ramped normal force obtained from a single scratch of these surface films are given in Fig. 6a and b for thin films of Co/CoW$_l$ and CoW$_m$/CoW$_h$, respectively. The progressive increase in the applied normal force results in a gradual increase in the force that acts in the direction parallel to the coating surfaces, known as the lateral force. When the scratch just penetrates any of these four films, a sharp change in the friction coefficient occurs. This moment is associated with fluctuations in the lateral force, used to define the critical applied load for film delamination. Due to the subtle nature of fluctuations in the lateral forces responsible for film delamination, the black straight lines in Fig. 6a and b serve as the fitting curves which facilitate the identification of the critical applied loads for the four surface films. Here, three meaningful scratches were made for each surface film, from which the average critical load with a standard deviation for film delamination was obtained, As shown in the bar graph of Fig. 6c, the critical loads of the Co, CoW$_l$, CoW$_m$ and CoW$_h$ films are 1466 ± 134 μN, 1811 ± 99 μN, 2266 ± 96 μN and 2115 ± 165 μN, respectively.

The critical load corresponds to the applied normal force at which the surface film just delaminates from its supporting substrate. Here, the adhesion strength (σ_c) of the film can be calculated from the previously reported mechanical-data related equation:

$$\sigma_c = \left(2P_c / \pi d_c^{\,2}\right)\left[\left(4+\nu_f\right)3\pi\mu / 8 - \left(1-2\nu_f\right)\right] \quad (2)$$

[33,34], where P$_c$ is the film-delamination force (i.e., critical load), d$_c$ is the width of the associated scratch track obtainable from the panoramic image of the whole scratch, ν_f is the Poisson ratio of the coated film assuming to be the same as bulk Co, 0.31, and μ is the friction coefficient, defined as the ratio between the lateral force and the normal force [35]. By dividing the lateral forces by the normal forces in the force plots of Fig. 6a and b, the dynamic friction coefficients of the Co, CoW$_l$, CoW$_m$ and CoW$_h$ films at the critical loads are derived to be almost the same (0.29−0.30). By fitting the above measured data to Eq. (2), the adhesion strengths of the Co, CoW$_l$, CoW$_m$ and CoW$_h$ films are calculated to be 35.8 MPa, 43.4 MPa, 56.1 MPa and 52.7 MPa, respectively. The three tungsten additions all evidently improve the cobalt's adhesion strength. Among the three W-added surface films, the thin CoW$_m$ film has the highest adhesion strength (56.1 MPa), in sharp contrast to the much lower value of the unalloyed Co film (35.8 MPa).

The ability of a coating against penetration is indicative of its hardness. For the case of surface films, the hardness obtained from nanoscratch testing is often assessed by a comparative method [36]. Thus, the relative hardness of the Co and three different W-added Co films can be deduced from monitoring the variation of their normal displacements with the progressively increased normal forces, as shown in Fig. 6d. All four metallic films roughly show a straight line as the normal displacement exceeds 20 nm. Thus, the relative values of their hardness are procured from the slopes of the straight-line segments to be about 14.8, 15.6, 16.8 and 17.5 μN nm^{-1} for the Co, CoW$_l$, CoW$_m$ and CoW$_h$, respectively. It appears that the hardness of Co is increased with the increase of the W concentration — namely that the Co film is the

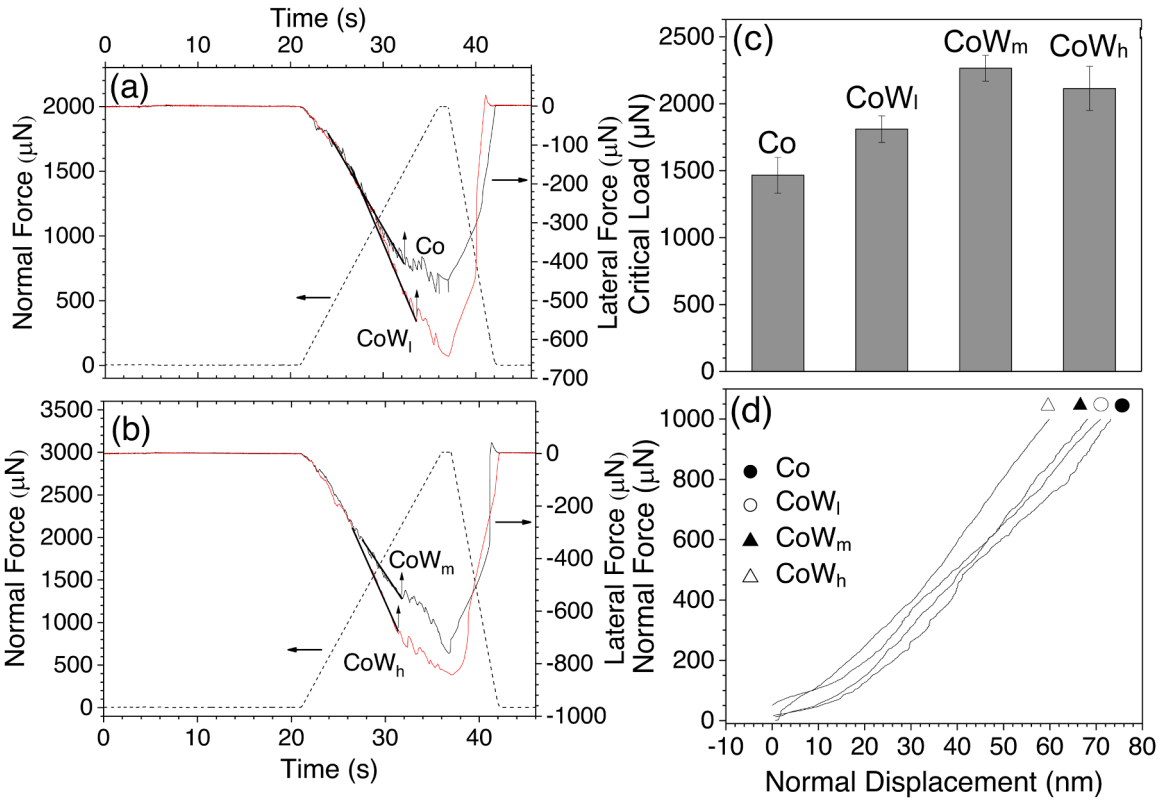


Fig. 6. Plots showing the variations of lateral force with ramped normal force for thin films of (a) Co/CoW_l and (b) CoW_m/CoW_h, together with (c) the associated bar graph of their critical loads causing film delamination and (d) normal force vs. normal displacement plots. The black straight lines in (a) and (b) function as fitting curves, enabling the identification of the critical applied loads.

softest material with the lowest hardness of 14.8 μN nm^{-1}; the CoW_m film has a markedly greater hardness of 16.8 μN nm^{-1}.

Surface energy includes dispersive and polar components due to van der Waals and polar interactions, respectively [37,38], which can be measured by the Owens-Wendt method [24]. Here, diiodomethane (nonpolar liquid) and water (polar liquid) with the established polar and dispersive surface-energy values, as reported before [39,40], were used separately for measuring the contact angles of their liquid droplets on the SAM-coated SiO_2 (substrate), thin Co and all W-added Co films. The measured surface energies and associated contact angles are summarized in Table 1, showing that the SAM/SiO_2 substrate has the highest specific surface energy of 74 mJ m^{-2}. Among the four metallic films, the Co film has the highest specific surface energy of 70 mJ m^{-2}, while the CoW_m film with the moderate W content (0.06 at.%) has the lowest specific surface energy of 60 mJ m^{-2}. However, increasing the W content from 0.06 at.% to 0.11 at.% results in a rise in the specific surface energy from 60 to 69 mJ m^{-2}; decreasing the W content from 0.06 at.% to 0.03 at.% also causes an increase in the specific surface energy from 60 to 66 mJ m^{-2}.

3.4. The correlation between thin-film properties and electromigration behaviors

The values of the adhesion strength, hardness, specific surface energy and electromigration mean time to failure (MTF) of Co, CoW_l, CoW_m and CoW_h, as obtained from the preceding sections, are compiled in Table 2 for comparison. According to this table, the extension of the lifetime and reduction in the scaling factor of Co by W alloying correlate closely with the improvement of surface and nanomechanical properties of the thin films. Notably, surface energy is an important thermodynamic factor that decides the wettability of a condensed matter to a heterogeneous supporting substrate. The deposited film with a lower surface energy would have a higher wetting tendency and thus should be with a greater substrate adherence than the counterpart with a higher surface energy. Indeed, the adhesion strength of the Co films after adding 0.03 at.% and 0.06 at.% of W is increased from 35.8 to 43.4 and 56.1 MPa, respectively, concurring well with the respective decrease in the surface energy from 70 to 66 and 60 mJ m^{-2} for CoW_l and CoW_m. As the added W is further increased to 0.11 at.%, the adhesion strength of the related CoW_h film is reduced to 52.7 MPa, associated with an

Table 1
The surface energies and associated contact angles for the sample films involved in this study. Note the standard derivations (about ± 2 %) were omitted in the table.

Sample Films	Contact Angle (°)		Surface Energy Data (mJ m^{-2})		
	H_2O	CH_2I_2	Polar	Dispersive	Total
SAM/SiO_2	8	5	40	34	74
Co	13	42	47	23	70
CoW_l	26	37	39	27	66
CoW_m	34	58	44	16	60
CoW_h	18	43	46	23	69

Table 2
The measured data of electromigration failure, surface energy and film mechanical properties of Co, CoW_l, CoW_m and CoW_h.

Measured Data	Sample Films			
	Co	CoW_l	CoW_m	CoW_h
Scaling Factor	0.50	0.43	0.33	0.42
MTF (s)*	5000	7330	11,010	9255
Adhesion (MPa)	35.8	43.4	56.1	52.7
Hardness (μN nm^{-1})	14.8	15.6	16.8	17.5
Surf. Energy (mJ m^{-2})	70	66	60	69

*Stressing condition: RT; $J = 5 \times 10^7$ A cm^{-2}.

increase in the surface energy to 69 mJ m^{-2}. The measured data of CoW_l and CoW_h (high tungsten content) obviously deviates from the inverse variation trend for Co, CoW_l and CoW_m. This deviation can be attributed to the strong affinity of W for oxygen [41], which likely results in a higher sticking coefficient of W atoms on the SiO_2 substrate compared to Co atoms [42]. This interfacial reaction likely facilitates the formation of tungsten oxide between the CoW_h and SiO_2, thereby enhancing adhesion strength. Such behavior has also been reported in Co-Zr, Co-Al [43] and Co-W-B [44] barriers for Cu metallization. The high concentration of W in CoW_h promotes the diffusion of W atoms and the subsequent formation of tungsten oxide. This process, coupled with the high surface energy of 69 mJ m^{-2}, results in a moderate adhesion energy of 52.7 MPa. Note that. The data of Table 2 apparently suggest that the magnitude of the adhesion strength, along with the film's hardness, is a key electromigration-mitigation factor, which in turn places the electromigration-reliability ranking of the four different Co deposits in decreasing order from CoW_m, CoW_h, CoW_l to Co.

Current flow through a conductor under the action of an electric field simultaneously places two forces on its atoms. One, the so-called direct force, is the electrostatic force exerted by the electric field on the ion cores, and another is the wind force due to the momentum transfer between current-charge carriers (vastly electrons with an elementary charge e) and the metal atoms [45]. As both forces are proportional to the electric field E and a valence, the net electromigration force F is conveniently related to the following equation:

$$F = (Z_{direct} + Z_{wind})eE = Z * eE, \tag{3}$$

where Z_{direct} is the nominal valence of the metal atoms (ignoring the dynamic screening effect), Z_{wind} is the electron-wind valence, and Z* stands for the effective valence. Electromigration involves diffusion of metallic ions driven by the electromigration force F. Upon electromigration stressing, the ion diffusion is typically via a vacancy-transfer mechanism with a flux J (in units of atoms $cm^{-2}.s^{-1}$) determined by the atomic diffusivity D, the vacancy concentration C, the operating temperature T, and the electromigration force F as:

$$J = DC/kT(Z*)eE \tag{4}$$

[46,47]. For the electrical stressing performed below half the absolute melting point of a stressed metal, electromigration mass transport is predominantly through grain boundaries (and interface) via a vacancy diffusion mechanism.

The forced motion of metal atoms along their jump path towards their neighboring vacancies leads to electromigration, eventually causing the vacancies around the grain boundaries to deviate from the equilibrium concentration. Assuming a negative effective valence, vacancies are depleted on the anode side and accumulated on the cathode side of the grain boundaries [48,49]. So, there is a vacancy concentration gradient decreasing from cathode to anode. This gives rise to the cathode side to have tensile forces and the anode side compressive forces. The internal stress gradient in turn triggers atomic migration opposing the electromigration atomic flux [31]. To account for this back stress effect, a diffusion term has been included [8,50], and the total electromigration flux is described as:

$$J(em) = -D\partial C/\partial x + DC/kT\ (Z^*)eE. \tag{5}$$

Our experiments were made at ambient temperature being well below half of the cobalt's absolute melting point. Thus, the mass transport of cobalt occurs primarily through the grain boundaries via a vacancy diffusion mechanism. If the equilibrium of the total atomic flux, as depicted in Eq. (5), is disturbed at triple points where grain boundaries branch off, by the metal/substrate interfaces with existing strong electron crowding effect, or by some inhomogeneous factors, such as localized change in grain sizes [31,51], a divergence in the atomic or vacancy flux will be present that creates extra vacancies for the nucleation of voids and subsequent growth of the voids in the surrounding damaged area.

Here, the content of the W in Co is minor (0.03, 0.06 or 0.11 at.%), and thus the effective valence Z*, diffusivity D and electrical resistivity ρ of the alloyed Co are assumed to be the same as those of the unalloyed Co. Stressing the Co and W-added Co lines with an identical current density under the same temperature should induce the same electromigration flux at equilibrium according to Eq. (5). Thus, the reason for the W-added Co lines to yield a higher resistance to a variety of flux divergences showing a reduced void growth rate and extended electromigration lifetime, as implicitly confirmed by their smaller n factors, is related to the higher film mechanical properties. As stated, compressive and tensile stresses tend to build up, respectively, in the anode and cathode sides of a damaged region around the grain boundaries or material interfaces. The relaxation of the stress gradient by a back-diffusion process accelerates the formation and growth of the voids there [52]. The high adhesion strength and hardness of the W-added Co, especially CoW_m, contribute to the inhibition of stress relaxation during electromigration stressing, thus mitigating electromigration and prolonging interconnect's lifetime. A thin polycrystalline conductive film with a higher adhesion strength and hardness would have a stronger substrate's binding force and a tighter (or tougher) grain structure, which would put it more resistant to electromigration voiding or thermal stressing [53,54]. Indeed, the SEM micrographs in Fig. 3 show that the thermal stability of the Co film is worse than that of the CoW_m film; the Co interconnect line also has an inferior electromigration reliability, as compared to all W-added Co interconnect lines.

3.5. Failure-profile and grain-structure analyses

Grain structure is a crucial factor in defining the strength or weakness of thin solid films. Most noticeably, among the four thin-film materials, CoW_m (W = 0.06 at.%) and Co have the best and worst electromigration reliability, respectively. Here, TEM/STEM was used to analyze two narrow lines of CoW_m and Co (the reference sample) after electromigration stressing for a given time at 5.0×10^7 A cm^{-2} and ambient temperature. The MTFs of Co and CoW_m lines were about 5000 s and 11,000 s, respectively. Here, the stressing times of the two Co and CoW_m lines were preset at half of the failure times as 2500 and 5500 s, respectively. Bright-field TEM images were first recorded over broad regions of the two lines along their longitudinal sections, as shown in Fig. 7a and b. It can be seen from Fig. 7a that electrical stressing of the Co line for 2500 s has induced the formation of many dome-shaped voids about 250 nm long and 50−80 nm deep which are located predominantly along the interfaces of the metal line with the surrounding substrate. This situation suggests that electromigration flux divergences of the Co atoms indeed occur at the grain boundaries and interfaces. As shown in the inset of Fig. 7a, prolonged electromigration stressing for 4200 s causes the voids to grow, spreading themselves transversely into open slits with 300−500 nm in length. Subsequently, the line is ruptured at 5000 s. Fig. 7b shows, however, that stressing the CoW_m line for a much longer time of 5500 s only causes the formation of much smaller voids without the formation of any open slits. The imaging analysis suggests again that CoW_m has a better electromigration resistance than Co. It should be noted that the voids in the Co interconnect line are located predominantly along its interfaces with the underlying substrate. It can be inferred from the void location and wear-out profile evolution that, upon electromigration stressing, the depletion of the Co atoms by grain-boundary diffusion starts from the metal/SiO_2 interfaces where there probably existing the strongest electron crowding effect and atomic flux divergence [36]. Moreover, the dome-shaped voids display a larger length than depth, suggesting preferential electromigration paths for the Co and W-added Co lines.

Fig. 7a and b suggest that adding 0.06 at.% of W to interconnect lines of Co results in a greatly reduced void growth rate, associated with a reduction of the 0.50 scaling factor of pure Co to 0.33. Closely comparing the bright-field TEM micrographs in Fig. 7a, b seems to

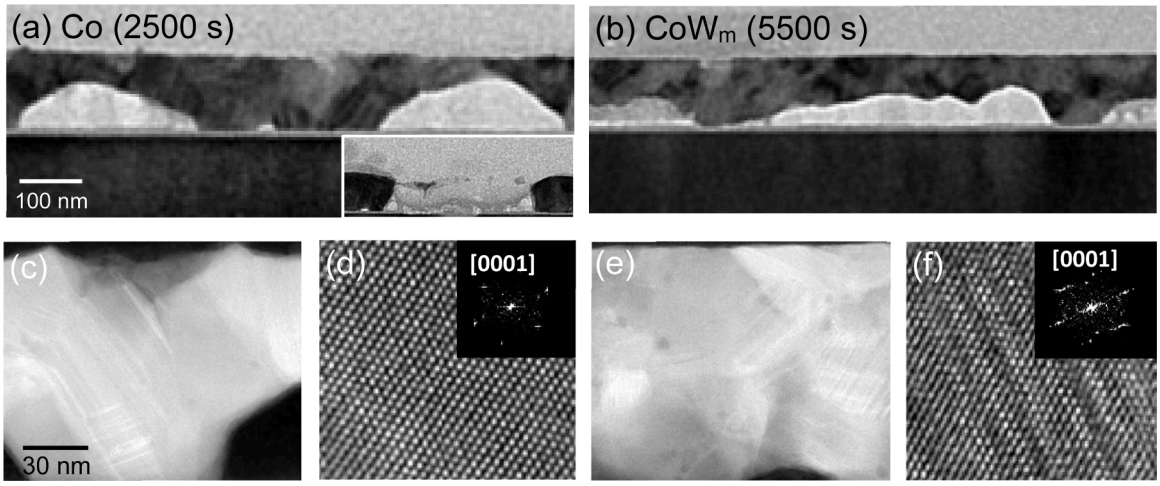


Fig. 7. Bright-field TEM images, taken from longitudinal sections of (a) Co and (b) CoW_m interconnect lines, showing their failure profiles after bias stressing for 2500 and 5500 s, respectively. Inset of (a) showing the Co-line failure profile and the cobalt accumulation on the top-free surface after 4200 s stressing. The associated HAADF and lattice-fringe images respectively in (c), (d) for Co and (e), (f) for CoW_m, along with their [0001]-oriented Fourier transform patterns (insets), revealing their grain-structure difference.

reveal the existence of a significant grain-structure difference between the Co and CoW_m lines. This difference was further analyzed by STEM-HAADF (high-angle annular dark-field) imaging and high-resolution TEM, as shown in Fig. 7c, d (Co) and Fig. 7e, f (CoW_m). The STEM-HAADF image in Fig. 7c shows that the Co line is made of large, columnar crystalline grains. It appears that some grains span the height of the entire interconnect line. The incoming mass flux reaching such 'blocking' grains tends to induce a flux divergence that is readily to move upwards along their 'straight' grain boundaries to the top free surface, later leading to the formation of large voids in the vicinity of these grains. As seen from the micrograph inserted in Fig. 7a, stressing for 4200 s causes Co atoms to migrate along the interfaces, free surface and grain boundaries, which then accumulate on the top free surface. Eventually, electrical discontinuity in the conductor line leads to open-circuit failure.

The STEM-HAADF image of the unalloyed Co line (Fig. 7c) exhibits a 'clean, smooth' contour, suggesting a relatively low defect density; the line also gives a 'perfect' lattice fringe image (Fig. 7d) and [0001] zone-axis Fourier transform pattern of HCP-Co (inset, Fig. 7d). By contrast, the CoW_m line comprises a stacking of equiaxed grains (Fig. 7b), seemingly offering a more complicated grain structure that could inhibit grain-boundary electromigration. Moreover, the CoW_m line exhibits an HAADF image with an intricate, rough contour (Fig. 7e), suggesting a relatively high defect density. Its high-resolution TEM image (Fig. 7f), taken along the [0001] zone axis, clearly shows the existence of stacking faults on (0001) basal planes, associated with a streaky diffraction pattern of HCP-Co (inset of Fig. 7f). Comparing the unalloyed Co, the equiaxed CoW_m has a more complex grain structure and a higher density of immobile planar defects, which would offer a maze of diffusion routes to retard the void-growth rate.

It should be noted that an STEM-HAADF image is produced by collecting the high-angle scattered electrons due to Rutherford scattering of atomic nuclei using an annulus detector around the principal axis of a primary incident electron beam. The collected signals are thus highly sensitive to the variation in the atomic number of the atoms in the imaged specimen, as opposed to the conventional TEM dark-field imaging method which utilizes extremely small-angle Bragg diffracted electrons. The high dependence of imaging contrast on atomic number makes STEM-HAADF a facile way to resolve a higher atomic-number alloy from its matrix material, as for the case of the W in Co matrix. However, using STEM-HAADF imaging, high-resolution TEM and STEM-EDX line scanning/mapping turned out to be inconclusive regarding the location of the added W, indicating that 0.06 at.% of W could be beyond the detection limit of the Talos™ F200X microscope.

In sum, the mitigation of Co electromigration by the alloying of extremely diluted W is directly related to an improvement of the film's mechanical properties and a reduction in the film's surface energy. Grain-structure refinement also plays a role in the reliability reinforcement.

4. Conclusions

An all-wet SAM-seeding and electroless-plating process has been developed to form thin films of Co and extremely diluted CoW alloys with controlled thin-film properties, including deposition rates, alloy compositions, deposit surface morphologies and microstructure stability against thermal stressing. We then extend this all-wet process to fill these metal films into SiO_2 trenches, eventually fabricating nanolines of three different CoW alloys for electromigration-behavior evaluations, using nanolines of Co as a reference. The electromigration mitigation for all three nanolines of Co alloys with 0.03 at.%, 0.06 at.% and 0.11 at.% of added W, termed CoW_l, CoW_m and CoW_h, respectively, under accelerated current stressing conditions of the order of 10^7 A cm^{-2} has been confirmed. Most noticeably, the CoW_m line containing a moderate amount of W, while only 0.06 at.%, yields a significantly reduced current density scaling factor of 0.33, which is much smaller than that of the Co line (0.50). So, it also yields a much slower void growth rate, associated with an electromigration lifetime at least twice as long as that of the Co line. While a variety of advanced TEM analyses fail to locate the trace amount of W which is less than one atom in about 1000 cobalt-matrix atoms, the 0.06 at% added W markedly elevates the adhesion strength and hardness of Co from 35.8 to 56.1 MPa and 14.8 to 17.5 μN nm^{-1}, respectively. Moreover, it also reduces Co surface energy from 70 to 60 mJ m^{-2}, favoring the wetting of the electroless deposits onto the underlying substrate. A fourth benefit given by the optimum W content, as found from advanced STEM/TEM analysis, is that the big electromigration-accelerating blocking grains, as exist in the Co line, are absent from the CoW_m line. The full account of the thin-film properties being investigated correlates well with the reliability performance of the four different metallic thin films/interconnects under thermal/electric-field stressing. It also reasonably explains why the ranking of their electromigration reliability being increased from Co, CoW_l (0.03 at.%), CoW_h (0.11 at.%) to CoW_m (0.06 at.%).

CRediT authorship contribution statement

Jau-Shiung Fang: Resources, Methodology. **Ting-Hsun Su:** Investigation, Data curation. **Yi-Lung Cheng:** Resources, Methodology. **Chun-Wei Huang:** Software, Data curation. **Giin-Shan Chen:** Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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